

Heaps /priority queue (introduction)
When we have to sort the data according to priority

Heaps & Priority Queue (Introduction)

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Heaps / Priority Queue

$O(n \log n) \Rightarrow$ sorting
 $O(n + k \log n)$ PQ

A $\rightarrow$ 3000
B $\rightarrow$ 5000
C $\rightarrow$ 10
D $\rightarrow$ 1000
E $\rightarrow$ 500

k=5
total=7

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05:03 / 07:25

Upaar wala elment highest priority wala hota hai priority queue me

Heaps & Priority Queue (Introduction)

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Heaps / Priority Queue

PQ

10
5
9
3
2
1

top

9, 10, 5, 3, 2, 1

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07:15 / 07:25

PQ IN STL

PQ in C++ STL

https://www.apnacollege.in/path-play...

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#include <queue>

PQ in STL

priority_queue<int> pq;

- push() //O(logn)
- pop() //O(logn)
- top() //O(1)

max heap
min heap
largest
smallest
Priority
Priority

02:24 / 08:04

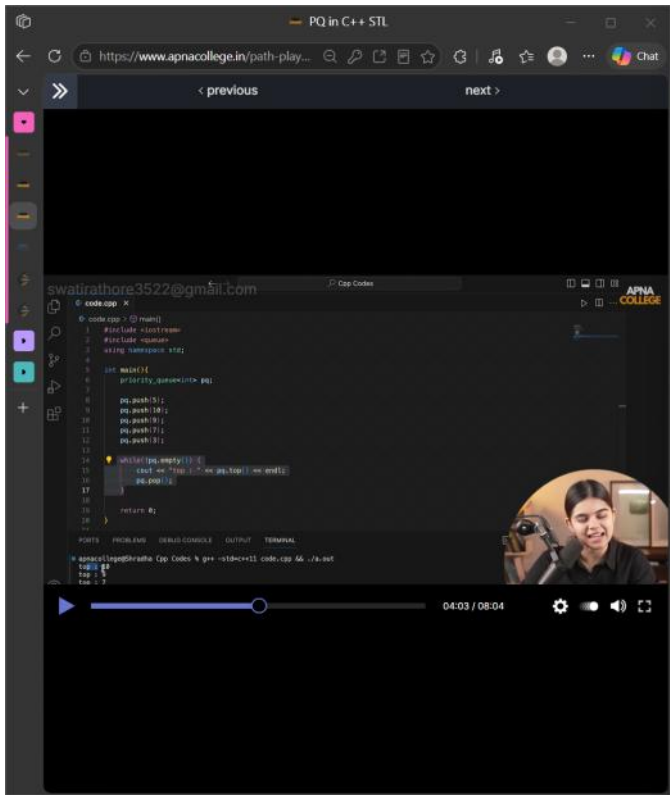
PQ in C++ STL

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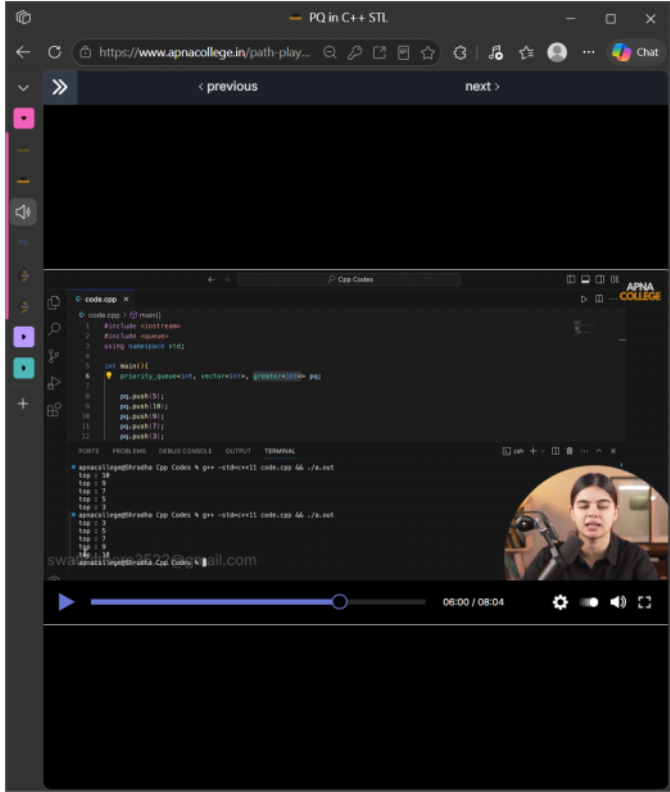
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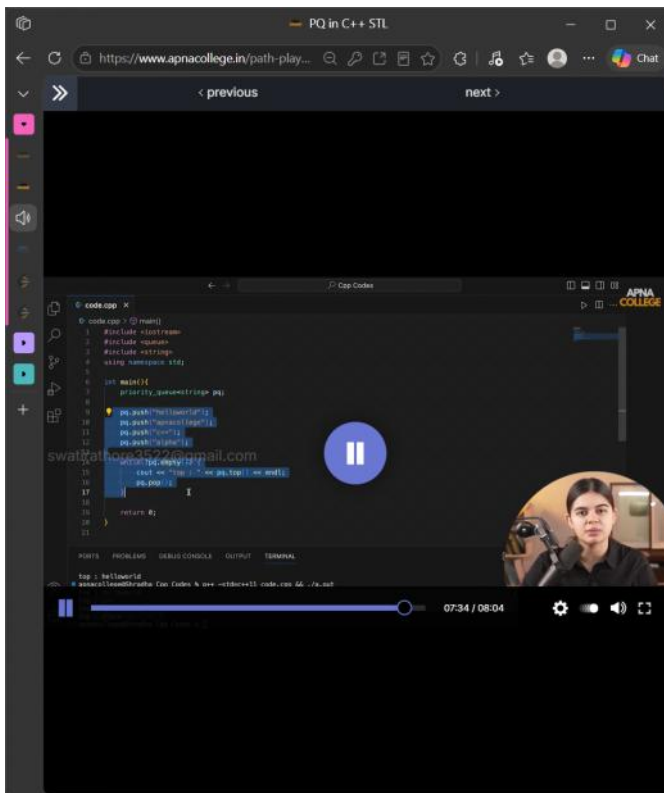
```
1 #include <iostream>
2 #include <queue>
3 using namespace std;
4
5 int main()
6 {
7     priority_queue<int> pq;
8     pq.push(5);
9     return 0;
10 }
```

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Ye to maxheap ho gaya
Ab minheap banaynge

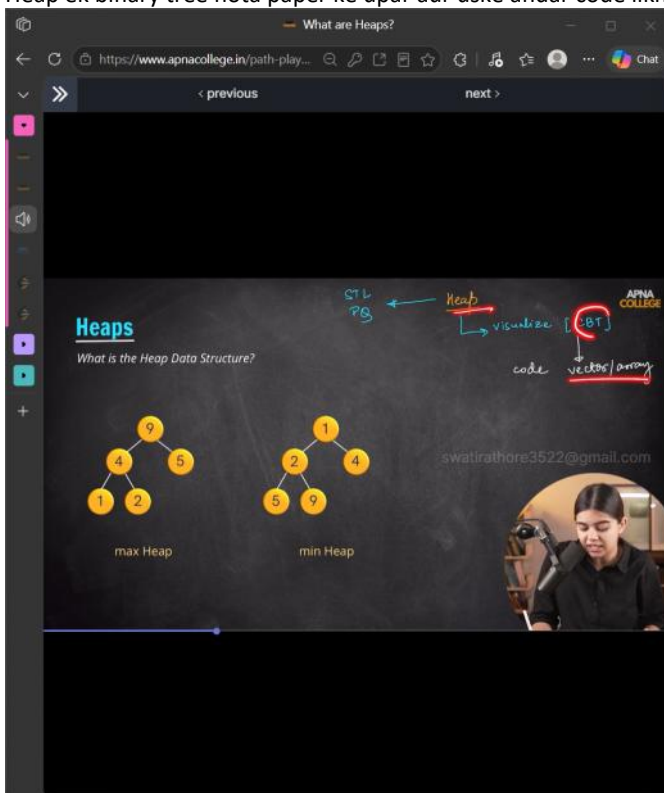




Iss wale me hum string input kar rahe

WHAT ARE HEAPS?

Heap ek binary tree hota paper ke upar aur uske andar code likha jata as a vector or array aur priority queue bani hoti hai heaps se



By default max heap create hota hai.

Percentage-maxheap

Rank-minheap

Left se right hi bhart ahi remember

What are Heaps?

Heaps

Heap is a Complete BT (CBT)

- CBT is a BT where all levels are filled except maybe the last one, which is filled from left to right
- Parent $\geq$ Children //Max Heap



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What are Heaps?

Heaps

Heap is a Complete BT (CBT)

- CBT is a BT where all levels are filled except maybe the last one, which is filled from left to right
- Parent $\geq$ Children //Max Heap



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06:02 / 10:01

Parent $\leq$ children minheap

What are Heaps?

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Which is a CBT?

Now maxheap check

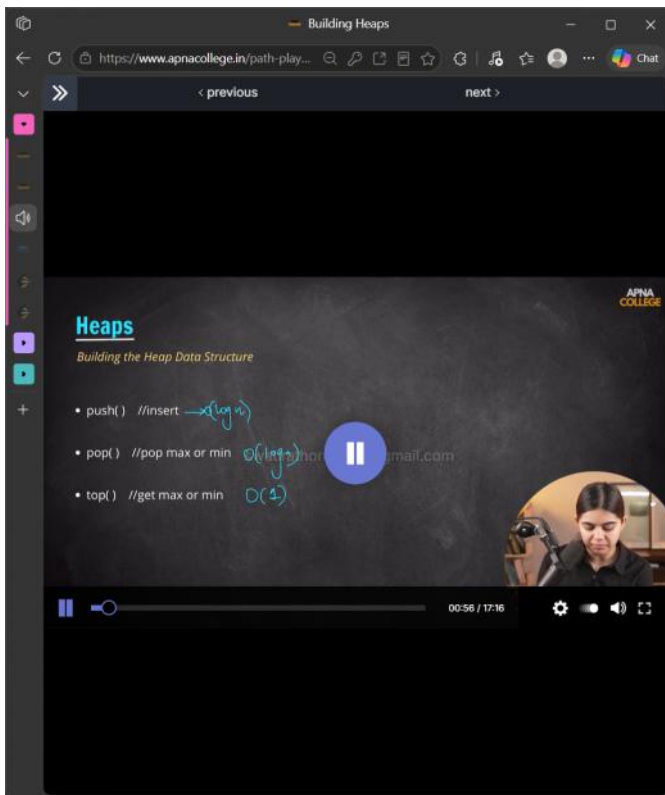
What are Heaps?

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Which is a Heap?

BUILDING HEAPS



Class me time complexity jayda lagti hai

Because

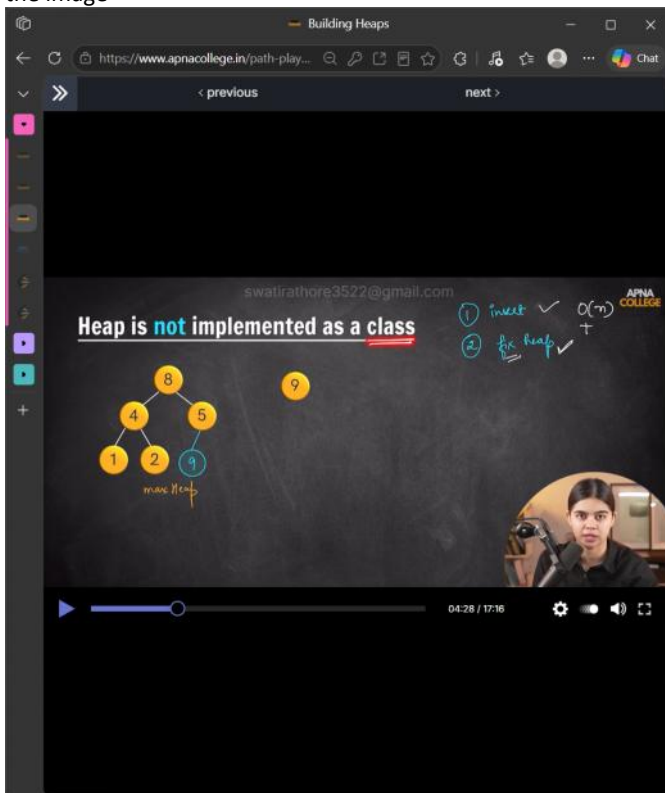
Nayi node insert karne ke liye

1-insert

2-fix heap

Normal binary tree ki tareh insertion me time lagega $O(n)$ aur phir additional time in fixing the heap isliye nahi class use nahi karte hai balki array use karte hai

Kyu kyuki heap ke andar hoti hai parent child relation sabse important property jo ki array/vector me rakha ja sakta hai kaise rakha ja sakta hai in the image



Building Heaps

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Heap as an array/vector

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Index i (parent)

left child $= 2i+1$
right child $= 2i+2$

child idx = x
par. =

10:32 / 17:16

Building Heaps

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Heap as an array/vector

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Index i (parent)

left child $= 2i+1$
right child $= 2i+2$

child idx = x
par. idx = $(x-1)/2$

11:08 / 17:16

Building Heaps

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Heap as an array/vector

Index i (parent) APNA COLLEGE

Left child = $2i+1$
Right child = $2i+2$

Child idx = 2
Parent idx = $(2-1)/2$

Index i (parent) APNA COLLEGE

Left child = $2i+1$
Right child = $2i+2$

Child idx = 2
Parent idx = $(2-1)/2$

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13:42 / 17:16

Personally 0 based indexing preferred by instructor

Array/vector ke last me insert karna sirf pushback karna hota vector me aur fix ke liye $O(\log n)$ use karte.

Building Heaps

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Heap as an array/vector

Index i (parent) APNA COLLEGE

Left child = $2i+1$
Right child = $2i+2$

Child idx = 2
Parent idx = $(2-1)/2$

Index i (parent) APNA COLLEGE

Left child = $2i+1$
Right child = $2i+2$

Child idx = 2
Parent idx = $(2-1)/2$

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PUSH IN HEAP

- 1-heap me insert
- 2- Heap ko fix

Push in Heap

Insert val = 9

max Heap

1 heap $\rightarrow$ insert(val)
vec.pb(val) $O(1)$

2 fix heap
P-C

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Push in Heap

Insert val = 9

max Heap

1 heap $\rightarrow$ insert(val)
vec.pb(val) $O(1)$

2 fix heap
P-C
child $I = 5$ (n)
parent $I = 2$ $(x-1/2)$

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Index compare hoge values nahi

Push in Heap

Insert val = 9

child = 5
parI = 2

max Heap

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while(child > parI){
 swap(child, parI)
 child = parI
 parI = (child - 1) / 2
}

1. Heap → sorted array
 vec: push_back

2. fix Heap
 child = 5 (n)
 parI = 2 (n/2)

Jab tak parent ke pass badi value hai tab tak swap karte raho aur phir we have to learn the time complexity $O(\log n)$ Of fixing

PUSH IN HEAP (CODE)

Push in Heap (Code)

```

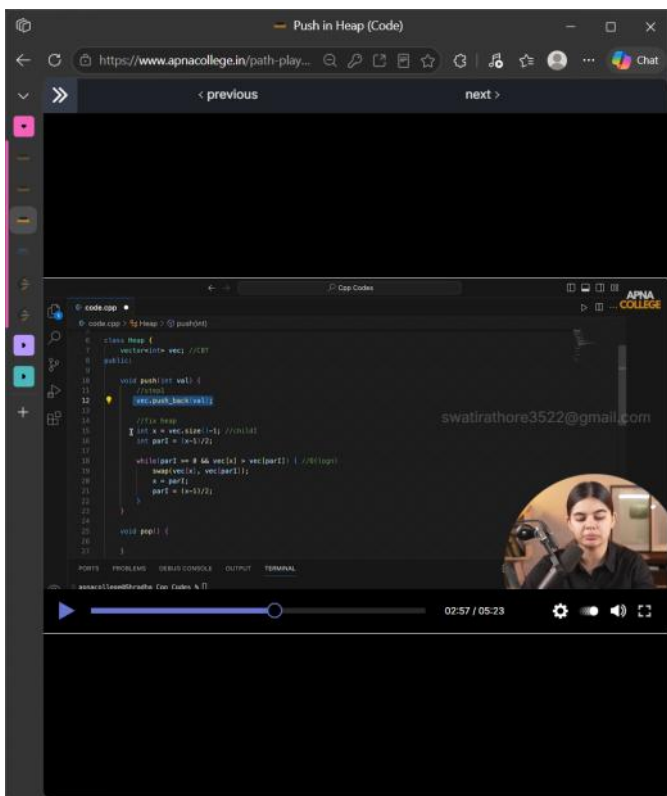
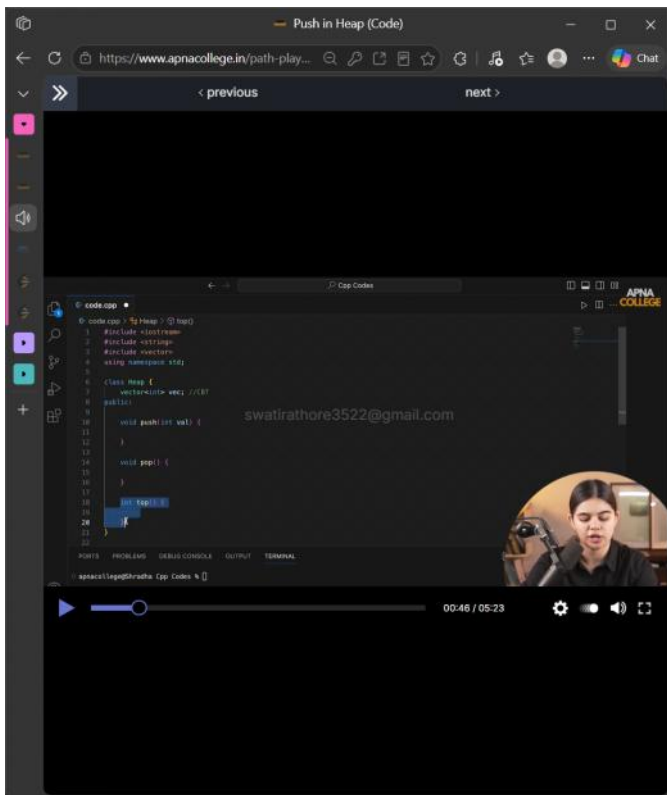
1 #include <iostream>
2 #include <string>
3 #include <vector>
4 using namespace std;
5
6 int main()
7 {
8     return 0;
9 }

```

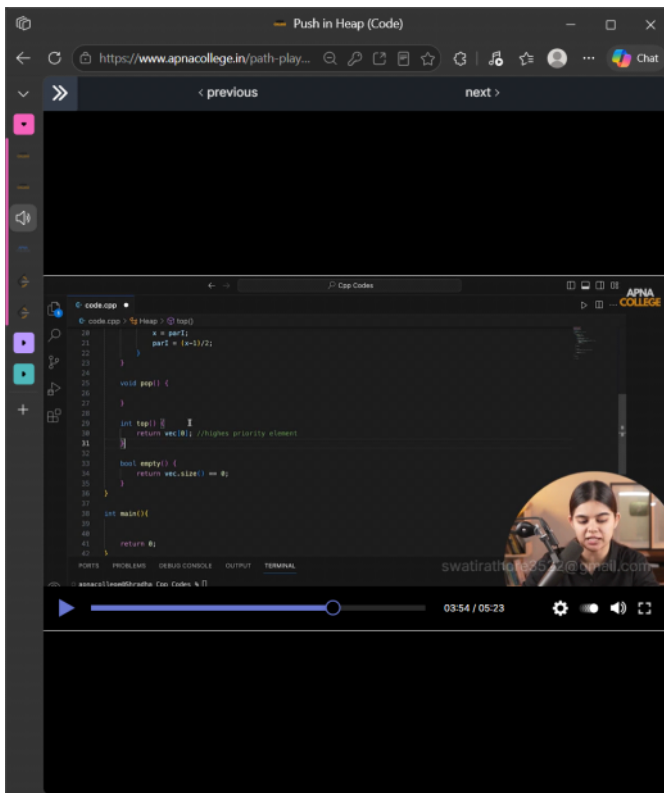
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00:05 / 05:23

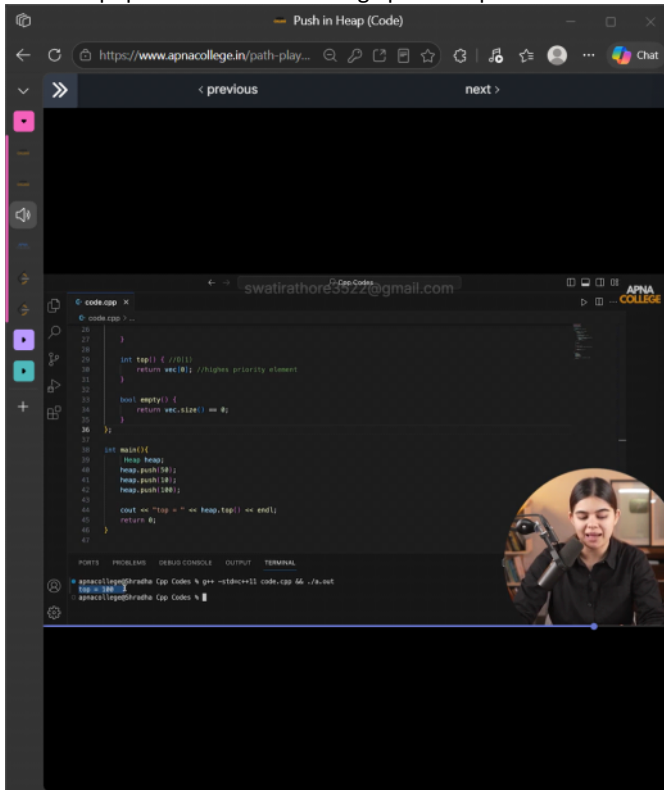
Kya kya banana



For boolean agr vector ka size zero to phir empty



Now the pop function wo bada hoga par usse pehle main me heap create karege



Min heap ke liye sari condition swap kar do ...

POP IN HEAD

Pop in Heap

1 swap (root, last) →

2 delete this lastIdx

3 fix the heap

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9 4 8 1 2 5

[9, 4, 8, 1, 2, 5]

0 1 2 3 4 5

Pop in Heap

1 swap (root, last) →

2 delete this lastIdx
rec. pop-back() → $O(1)$

3 fix the heap

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9 4 8 1 2 5

[9, 4, 8, 1, 2, 5]

0 1 2 3 4 5

Now we will fix the heap kyuki ab wo nahi hai fixed

Pop in Heap

1 swap (root, last) → O(1)

2 delete this lastIdx
rec.pop_back() → O(1)

3 fix the heap → O(log n)
heapify

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maxHeap

[9, 4, 8, 1, 2, 5]

0 1 2 3 4 5

03:34 / 13:03

Heapify (O(log n))

Heapify

O(log n)

heapify (idx = 0) {

val1 = arr[idx]

val2 = arr[2 * idx + 1]

val3 = arr[2 * idx + 2]

i = max(val1, val2, val3)

swap(arr[idx], arr[i])

}

maxHeap

[3, 8, 6, 4]

0 1 2 3

Teeno ka max nikala aur usse root par le gaye
Min ke liye bas min nikalega instead of max

Pop in Heap

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Heapify

$O(\log n)$

maxHeap

[3, 8, 6, 4]

$i = 1$
 $l = 2 \times i + 1 = 3$
 $r = 2 \times i + 2 = 4$

$heapify(i, arr, n)$
 $val1 \leftarrow arr[i]$
 $val2 \leftarrow l = 2 \times i + 1$
 $val3 \leftarrow r = 2 \times i + 2$
 $i \leftarrow \max(val1, val2, val3)$
 $swap(arr[i], arr[i])$
 $heapify(i, arr, n)$

[3, 8, 6, 4]

Worst case heapify function $O(\log n)$ because of that pop is $O(\log n)$

Pop in Heap

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Pop in Heap

maxHeap

[9, 4, 8, 1, 2, 5]

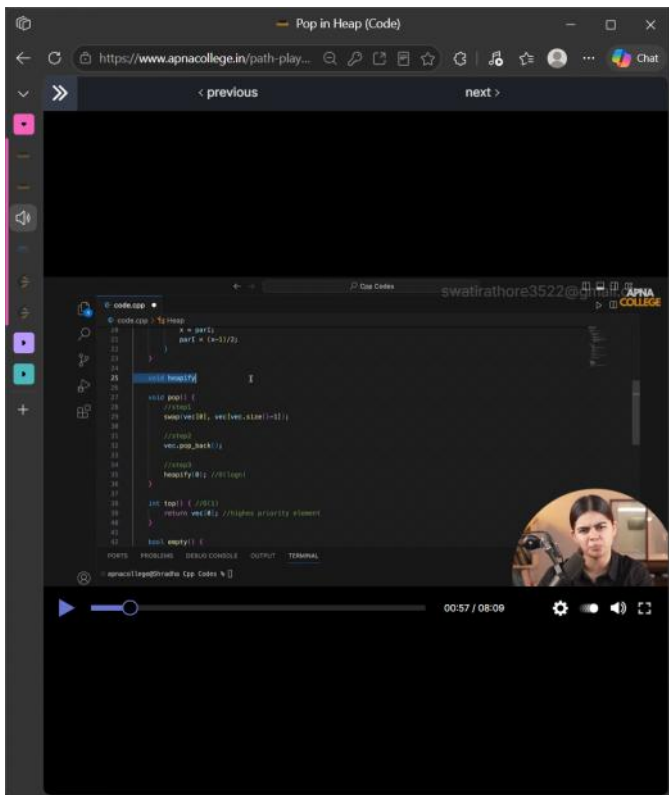
$swap(arr[0], arr[n-1])$
 $delete this last index$
 $rec.pop_back() \rightarrow O(1)$
 $fix the heap \rightarrow O(\log n)$
 $heapify \rightarrow O(\log n)$

$O(\log n)$

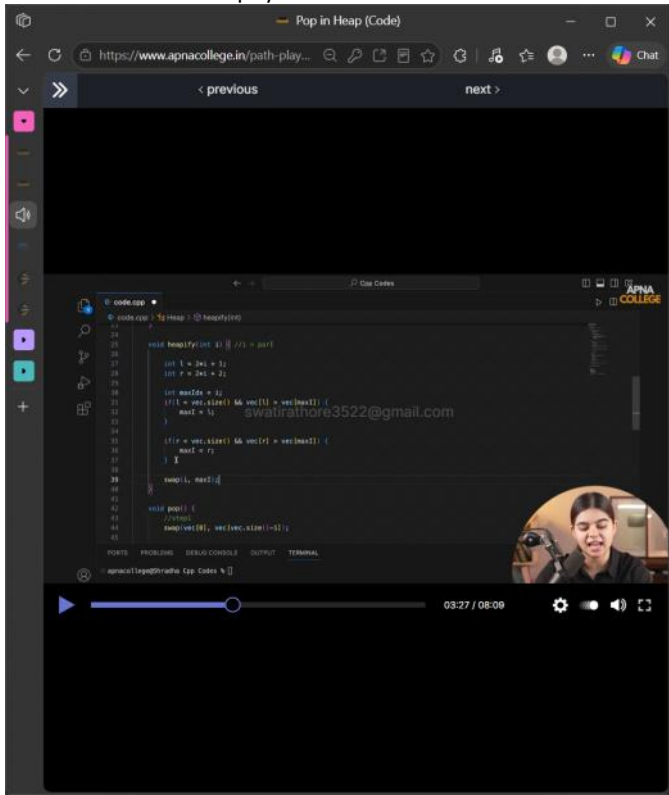
$1 \rightarrow swap(arr[0], arr[n-1])$
 $2 \rightarrow delete this last index$
 $3 \rightarrow fix the heap \rightarrow O(\log n)$
 $heapify \rightarrow O(\log n)$

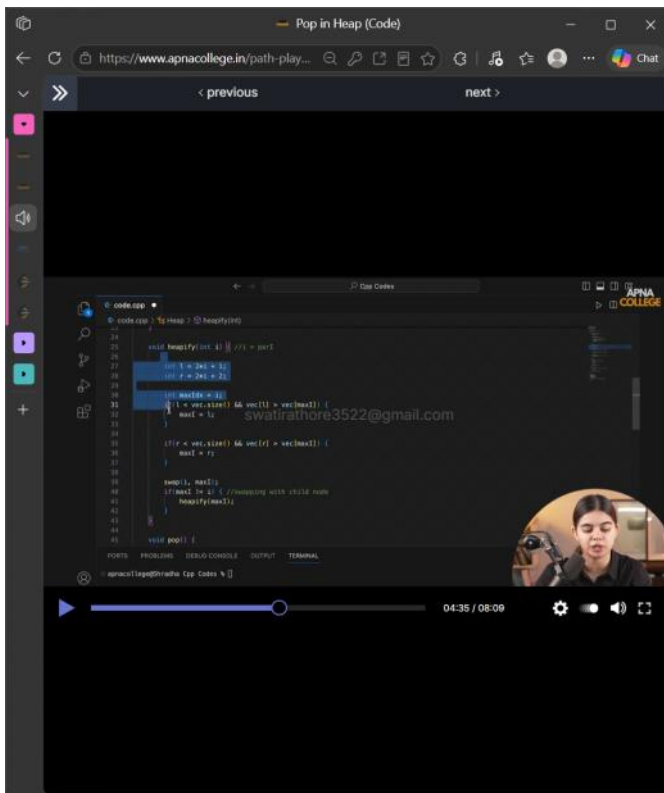
[9, 4, 8, 1, 2, 5]

POP IN HEAP(CODE)

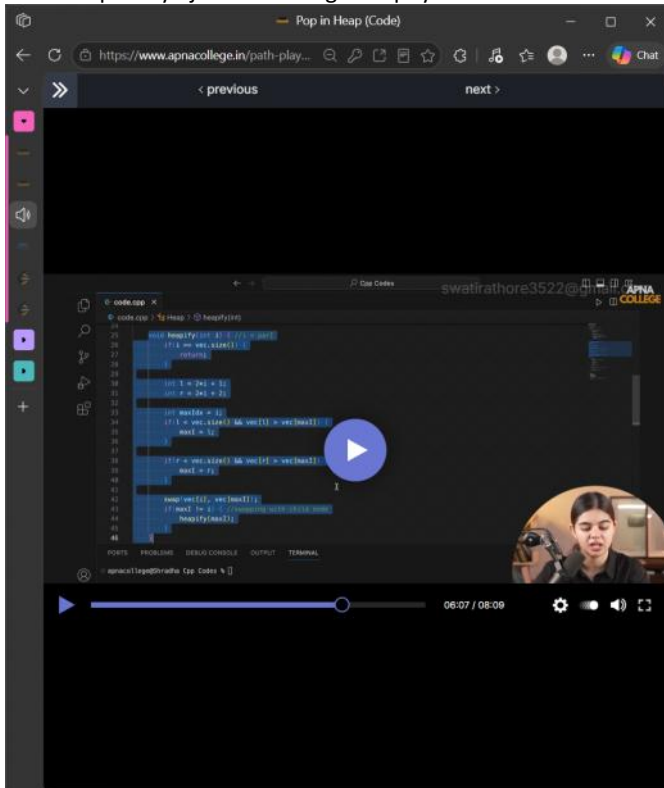


Now we will make heapify function

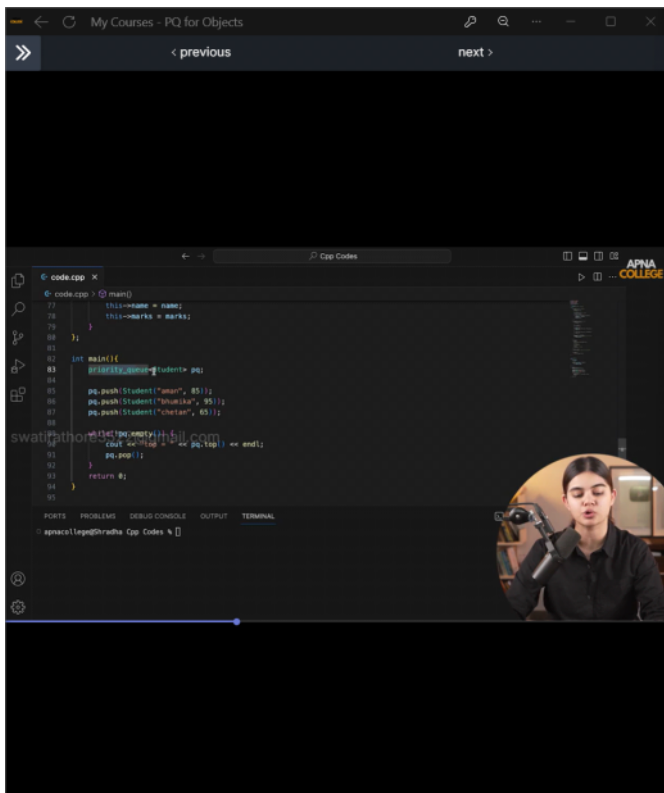




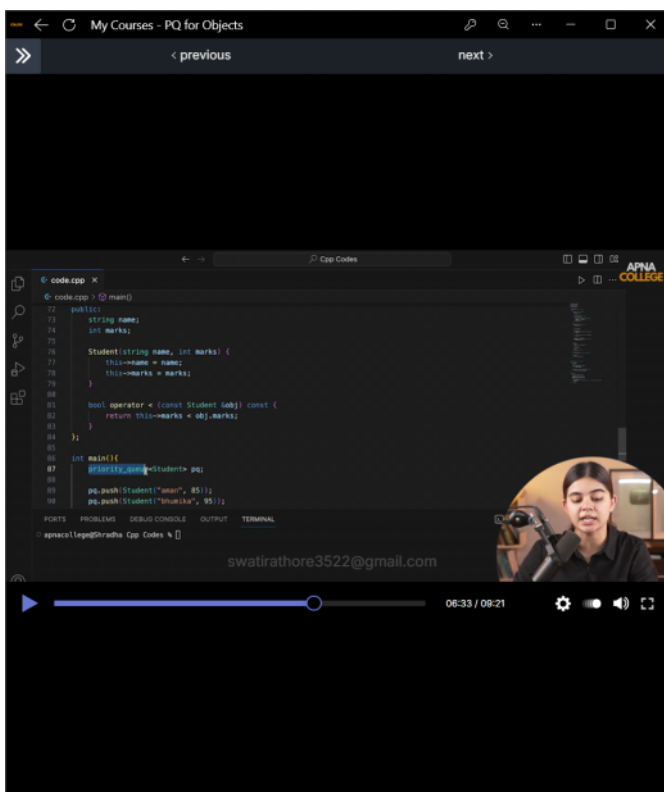
Agar values exchange karne ke baad neeche wala bigad jaye for that case we apply the if (maxi!=i) wala condition
Khali heap ke liye jab hum karege heapify to return kar den base case



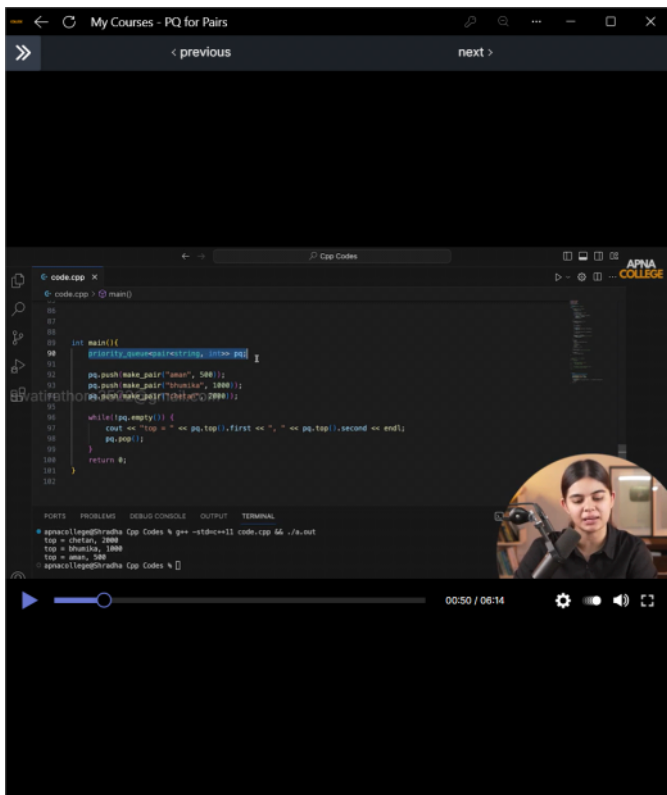
POP IN HEAP (DRY RUN)



Error aayga kyu konse object ke basis par heap build hona chahiye iske liye operator overloading



Marks ke basis par kiya abhi
Name ke basis par bhi kar sakte
Min heap ke liye sign reverse andar
PQ FOR PAIRS



Hum pairs ka struct bana lege
 Struct memory allocation stack ke andar karte heap memory allocation heap ke andar karte hai
 Struct for the pair comparison
 Neeche priority queue banate time main me vector pass karege phir struct pass

